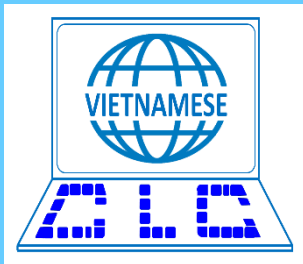


Section 3: Corpus Linguistics



Lecturer: Assoc.Prof. Dr. Dinh Dien

Introduction to Corpus

Definition: Corpus = “a collection of written or spoken texts”
(Oxford Dic)

“A corpus is a **collection** of **pieces** of language that are **selected** and **ordered** according to explicit **linguistic criteria** in order to be used as a **sample** of the language” (Sinclair 1996)

“[...] the term *corpus* as used in modern linguistics can best be defined as a collection of **sampled** texts, **written** or **spoken**, in **machine-readable form** which may be **annotated** with various forms of linguistic information” (McEnery, Xiao and Tono 2006)

- Translation of “corpus” = “语料库”/yǔ liào kù/(Chinese: ngữ liệu khố); “코퍼스”/kô-po-su/ (Ko); “コーパス” /kô-pa-zu/ (Jp); corpus (Fr), korpus (Ge), корпус (Ru),...

Criteria: representativeness, balance, sampling

Corpus Classification

Annotation: Raw (unannotated) vs. Annotated (linguistic information)

Annotating labels in different **aspects** and **linguistic units**

Aspects: phonetics, morphology, grammar, semantics, pragmatics

Linguistic units: phoneme, morpheme, morpho-syllable, word, phrase, sentence, paragraph, text.

Language: monolingual vs. Multilingual

Alignment: **parallel** vs. **comparable**

Prime Minister Pham Minh Chinh arrived in Kuala Lumpur on May 25, 2025:

Thủ tướng Phạm Minh Chính đã đến Kuala Lumpur vào ngày 25/5/2025.

Người đứng đầu chính phủ VN đã đến thủ đô Malaysia vào ngày hôm qua.

Parallel: text – paragraph – sentence – phrase - word alignment

Translation vs. **Transliteration** vs. **Interpretation**

Corpus samples

- **PTB (Penn Tree Bank):** [Pierre/NNP Vinken/NNP],/, [61/CD years/NNS] old/JJ ,/, will/MD join/VB [the/DT board/NN] as/IN [a/DT nonexecutive/JJ director/NN Nov./NNP 29/CD]/./.
- **CTB (Chinese Tree Bank):** <S ID=12>((IP-HLN (NP-SBJ (NN 外商) (NN 投资) (NN 企业)) (VP (VV 成为) (NP-OBJ (NP (NP-PN (NR 中国)) (NP (NN 外贸))) (ADJP (JJ 重要)) (NP (NN 增长点))))))) </S>
(VTB: Vietnamese Tree Bank): <SEG id="1">
Nguyễn_nhân/Nn/O là/Vc/O bảo/Nn/O số/Nn/O 10/An/O
đang/R/O chịu/Vv/O ảnh_hưởng/Nn/O bởi/Cp/O
hệ_thống/Nn/O trực/Nn/O rãnh/Nn/O cao/Aa/O và/Cp/O
sự/Nc/O lôi_kéo/Vv/O từ/Cm/O siêu__bảo/Nn/TRM_B
Melor/Nr/TRM_I ở/Cm/O ngoài/Cm/O khơi/Nn/O
Philippines/Nr/LOC_B ./PU/O</SEG>

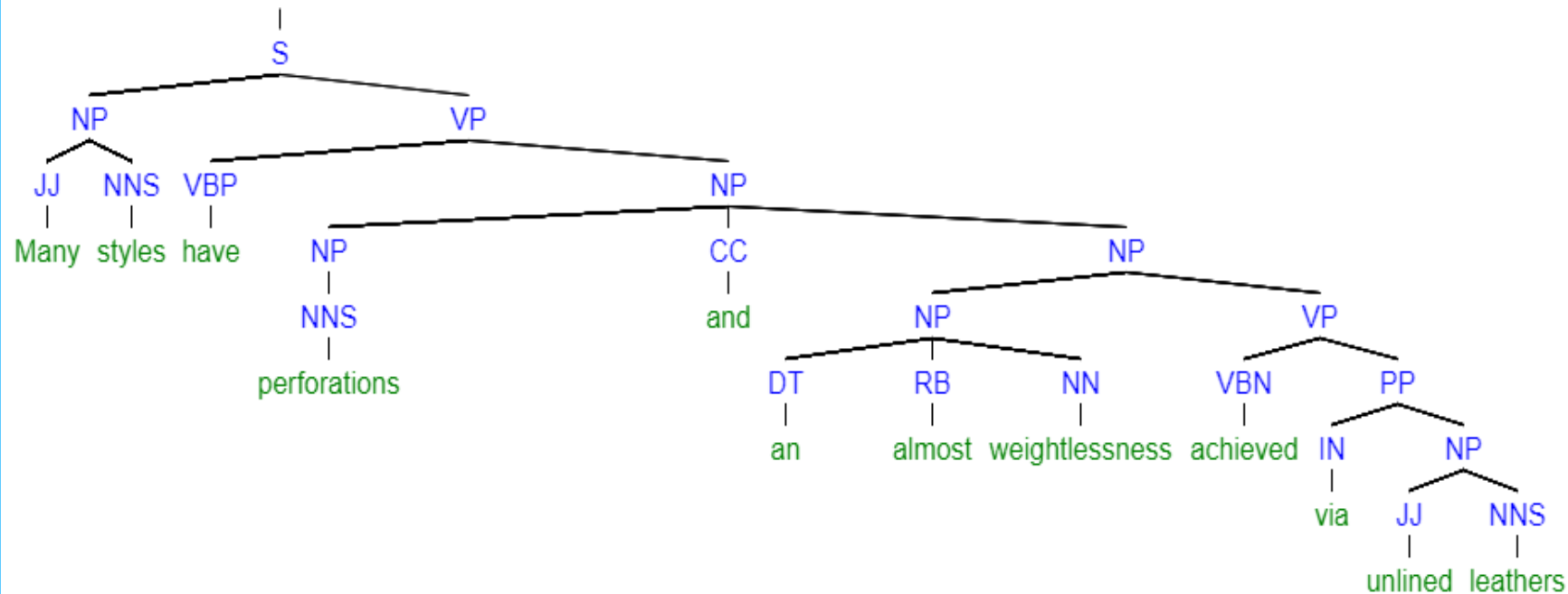
tagset

Tag	Description	Example	Tag	Description	Example
CC	Coordin. Conjunction	<i>and, but, or</i>	SYM	Symbol	<i>+, %, &</i>
CD	Cardinal number	<i>one, two, three</i>	TO	"to"	<i>to</i>
DT	Determiner	<i>a, the</i>	UH	Interjection	<i>ah, oops</i>
EX	Existential 'there'	<i>there</i>	VB	Verb, base form	<i>eat</i>
FW	Foreign word	<i>mea culpa</i>	VBD	Verb, past tense	<i>ate</i>
IN	Preposition/sub-conj	<i>of, in, by</i>	VBG	Verb, gerund	<i>eating</i>
JJ	Adjective	<i>yellow</i>	VCN	Verb, past participle	<i>eaten</i>
JJR	Adj., comparative	<i>bigger</i>	VBP	Verb, non-3sg pres	<i>eat</i>
JJS	Adj., superlative	<i>wildest</i>	VBZ	Verb, 3sg pres	<i>eats</i>
LS	List item marker	<i>1, 2, One</i>	WDT	Wh-determiner	<i>which, that</i>
MD	Modal	<i>can, should</i>	WP	Wh-pronoun	<i>what, who</i>
NN	Noun, sing. or mass	<i>llama</i>	WPS	Possessive wh-	<i>whose</i>
NNS	Noun, plural	<i>llamas</i>	WRB	Wh-adverb	<i>how, where</i>
NNP	Proper noun, singular	<i>IBM</i>	\$	Dollar sign	<i>\$</i>
NNPS	Proper noun, plural	<i>Carolinas</i>	#	Pound sign	<i>#</i>
PDT	Predeterminer	<i>all, both</i>	"	Left quote	<i>' or "</i>
POS	Possessive ending	<i>'s</i>	"	Right quote	<i>' or "</i>
PRP	Personal pronoun	<i>I, you, he</i>	(	Left parenthesis	<i>[, (, {, <</i>
PRP\$	Possessive pronoun	<i>your, one's</i>	)	Right parenthesis	<i>],), }, ></i>
RB	Adverb	<i>quickly, never</i>	,	Comma	<i>,</i>
RBR	Adverb, comparative	<i>faster</i>	.	Sentence-final punc	<i>. ! ?</i>
RBS	Adverb, superlative	<i>fastest</i>	:	Mid-sentence punc	<i>: ; ... --</i>
RP	Particle	<i>up, off</i>			

Figure 5.6 Penn Treebank part-of-speech tags (including punctuation).

[Many/JJ styles/NNS]
 have/VBP
 [perforations/NNS]
 and/CC
 [an/DT almost/RB weightlessness/NN]
 achieved/VBN via/IN
 [unlined/JJ leathers/NNS]
 ./.

```
( (S
  (NP (JJ Many) (NNS styles) )
  (VP (VBP have)
    (NP
      (NP (NNS perforations) )
      (CC and)
      (NP
        (NP (DT an) (RB almost) (NN weightlessness) )
        (VP (VBN achieved)
          (PP (IN via)
            (NP (JJ unlined) (NNS leathers) ))))))))
```



SUSANNE

A01:0010a	-	YB	<minbrk>	-	[Oh.Oh]
A01:0010b	-	AT	The the	[O[S[Nns:s.	
A01:0010c	-	NP1s	Fulton Fulton	[Nns.	
A01:0010d	-	NNL1cb	County county	.Nns]	
A01:0010e	-	JJ	Grand grand	.	
A01:0010f	-	NN1c	Jury jury	.Nns:s]	
A01:0010g	-	VVDv	said say	[Vd.Vd]	
A01:0010h	-	NPD1	Friday Friday	[Nns:t.Nns:t]	
A01:0010i	-	AT1	an an	[Fn:o[Ns:s.	
A01:0010j	-	NN1n	investigation investigation	.	
A01:0020a	-	IO	of of	[Po.	
A01:0020b	-	NP1t	Atlanta Atlanta	[Ns[G[Nns.Nns]	
A01:0020c	-	GG	+<apos>s	- .G]	
A01:0020d	-	JJ	recent recent	.	
A01:0020e	-	JJ	primary primary	.	
A01:0020f	-	NN1n	election election	.Ns] Po]Ns:s]	
A01:0020g	-	VVDv	produced produce	[Vd.Vd]	
A01:0020h	-	YIL	<ldquo> -	.	
A01:0020i	-	ATn	+no no	[Ns:o.	
A01:0020j	-	NN1u	evidence evidence	.	
A01:0020k	-	YIR	+<rdquo>	- .	
A01:0020m	-	CST	that that	[Fn.	
A01:0030a	-	DDy	any any	[Np:s.	
A01:0030b	-	NN2	irregularities irregularity	.Np:s]	
A01:0030c	-	VVDv	took take	[Vd.Vd]	
A01:0030d	-	NNL1c	place place	[Ns:o.Ns:o]Fn]Ns:o]Fn:o]S]	
A01:0030e	-	YF	+. -	.O]	
A01:0030f	-	YB	<minbrk>	- [Oh.Oh]	

Semantics: SEMCOR

```
kcontextfile concordance=brown>
<context filename=br-a01 paras=yes>
<p pnum=1>
<s snum=1>
<wf cmd=ignore pos=DT>The</wf>
<wf cmd=done rdf=group pos=NNP lemma=group wnsn=1 lexsns=1:03:00::
pn=group>Fulton_County_Grand_Jury</wf>
<wf cmd=done pos=VB lemma=say wnsn=1 lexsns=2:32:00::>said</wf>
<wf cmd=done pos=NN lemma=friday wnsn=1 lexsns=1:28:00::>Friday</wf>
<wf cmd=ignore pos=DT>an</wf>
<wf cmd=done pos=NN lemma=investigation wnsn=1 lexsns=1:09:00::>investigation</wf>
<wf cmd=ignore pos=IN>of</wf>
<wf cmd=done pos=NN lemma=atlanta wnsn=1 lexsns=1:15:00::>Atlanta</wf>
<wf cmd=ignore pos=POS>'s</wf>
<wf cmd=done pos=JJ lemma=recent wnsn=2 lexsns=5:00:00:past:00>recent</wf>
<wf cmd=done pos=NN lemma=primary_election wnsn=1
lexsns=1:04:00::>primary_election</wf>
<wf cmd=done pos=VB lemma=produce wnsn=4 lexsns=2:39:01::>produced</wf>
<punc>`</punc>
<wf cmd=ignore pos=DT>no</wf>
<wf cmd=done pos=NN lemma=evidence wnsn=1 lexsns=1:09:00::>evidence</wf>
<punc>'</punc>
<wf cmd=ignore pos=IN>that</wf>
<wf cmd=ignore pos=DT>any</wf>
<wf cmd=done pos=NN lemma=irregularity wnsn=1 lexsns=1:04:00::>irregularities</wf>
<wf cmd=done pos=VB lemma=take_place wnsn=1 lexsns=2:30:00::>took_place</wf>
<punc>.</punc>
</s>
</p>
```


Parallel corpus: alignment

Helicopters can rise straight up into the air and can go straight down. They can stand still in the air. Helicopters do not have wings. A huge whirling propeller, called a rotor, on top of a helicopter provides the lift.

Máy bay trực thăng có thể lên thẳng trên không và đáp thẳng xuống đất. Chúng có thể đứng yên trên không. Máy bay trực thăng không có cánh, một cánh quạt lớn gọi là chong chóng trên đầu chiếc máy bay cung cấp sức nâng.

* Helicopters can rise straight up into the air and can go straight down.

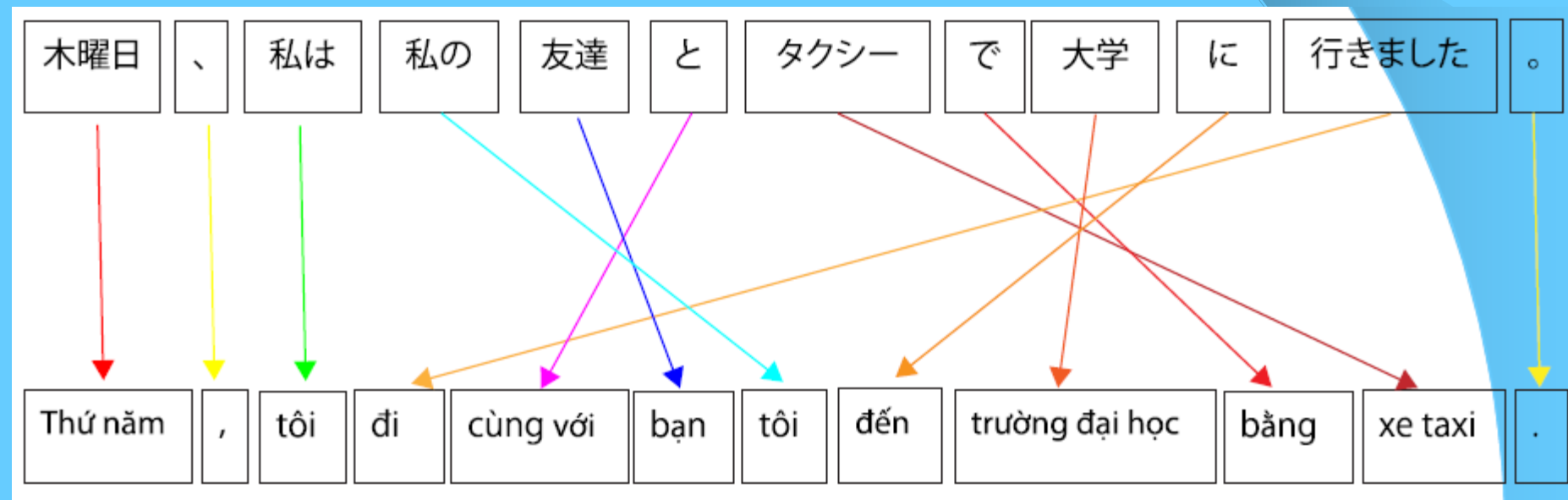
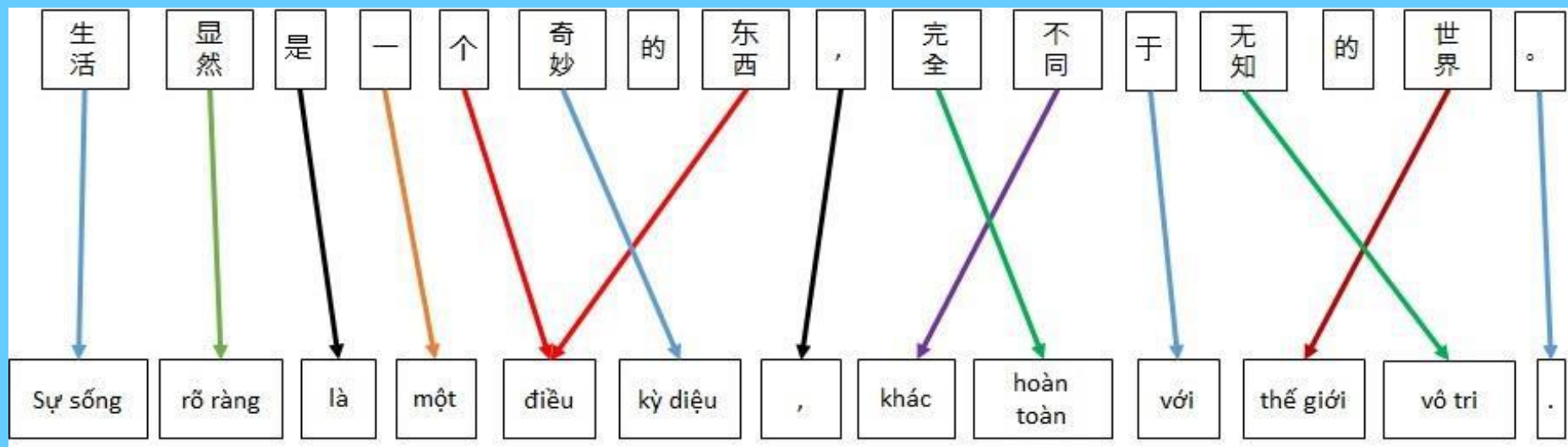
+ Máy bay trực thăng có thể lên thẳng trên không và đáp thẳng xuống đất.

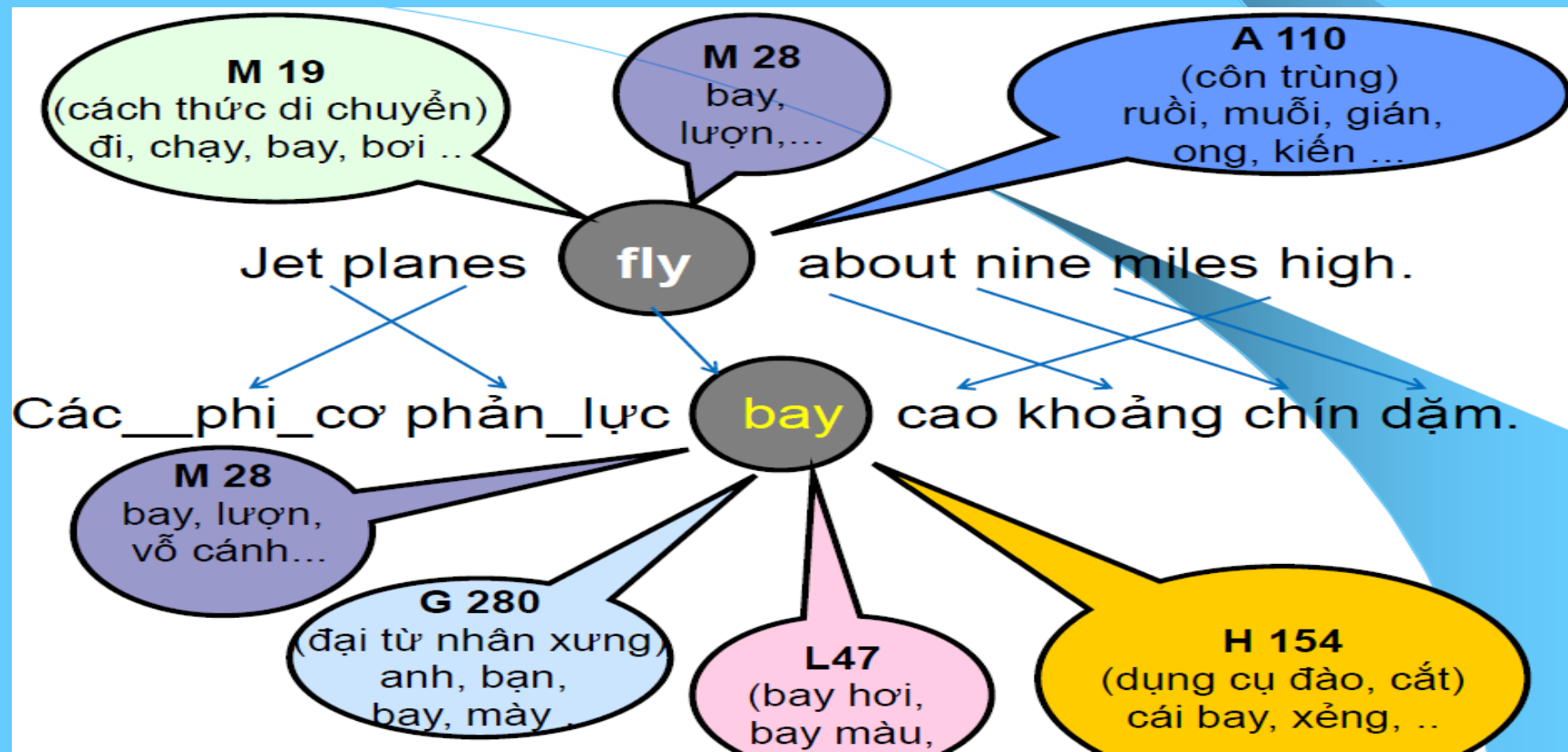
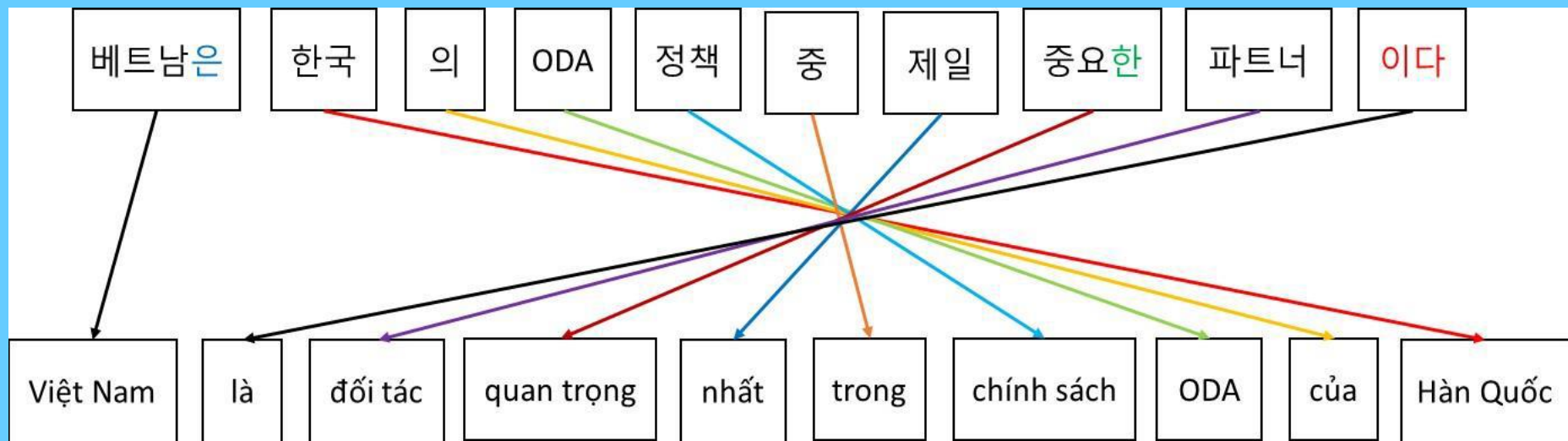
* They can stand still in the air.

+ Chúng có thể đứng yên trên không.

* Helicopters do not have wings.

+ Máy bay trực thăng không có cánh.





TRANSLITERATION

- ❑ Transliteration (chuyển tự): Replace letters from one form to another of the same language.
- ❑ Cyrillic (Russian) “Путин” into Latin “*Putin*”, similarly:
 - ✓ Hangul (Kr): 삼성 => *S-a-m S-o-ng* (tam tinh)
 - ✓ Greek: γραφικ => *graphic*
 - ✓ Hiragana (Jp): にほん => *ni-ho-n*
 - ✓ Hebrew: ירושלים => *Jerushalayim*
 - ✓ Arabic: الله => *Allah*
 - ✓ Thai: หม่า => *Hmā* (dog)
- Transliteration of the above phonetic script is very simple because there is an (almost) 1-1 correspondence between the letters of the two writing systems.
- However, converting Han Nom characters (meaning) into National Scripts (sound) is very difficult because of the phenomena of homonymy, heteronymy, polysemy, and not a 1-1 mapping.

Transliteration of Nôm scripts vs. National Scripts

Tale of Kieu

"Trời còn để có hôm nay
Tan sương đầu ngõ vén mây giữa trời"

天 群 底 固 歆 脛

Trời còn để có hôm nay,

散 霜 頭 午 援 逮 忡 天

Tan sương đầu **ngõ**, vén mây giữa trời.

nomfoundation.org/nom-tools/Nom-Lookup-Tool/Nom-Lookup-Tool?uiLang=vn



會保存遺產喃

English

[Thông tin về Hội](#) [Chữ Nôm](#) [Sách Nôm](#) [Từ điển Nôm](#) [Dự án Nôm](#) [Phông chữ Nôm](#) [Liên hệ](#)

Another Nôm Lookup Tool based on Unicode

- ☒ Quốc Ngữ hoặc Hán-Nôm ☐ Mã Unicode hoặc TCVN (dùng hệ hex) ☐ Tiếng anh
☐ Bắc kinh ☐ Quảng đông ☐ Thương hiệt
☐ Bộ thủ.Tổng nét

午

GO

TRA CỨU CHỮ NÔM

Tra cứu chữ Nôm

Câu chuyện về tra cứu Nôm

Quốc Ngữ	Hán-Nôm	Context	Ref.	Tiếng anh
ngõ	午	cửa ngõ	btcn	gateway
ngọ	午	giờ ngọ	vhn	midday, noon
ngỏ	午	ngỏ ý	btcn	to express a wish

- ✓ Chữ ghi âm: **thuận tiện** (dễ học), nhưng bị: **đồng âm, dị nghĩa**
- ✓ “ba”: số 3, ba mẹ, phong ba/ba đào (microwave), ba ba,
- ✓ “co”: cơ hàn (飢), cơ sở (基), cơ khí (机), cơ bắp (肌),...
- ✓ Chữ Hán: 味 => “mùi” (âm HV cổ) hay “vị” (từ đời nhà Đường)
- ✓ Chữ Hán/Nôm: “化” => “hóa; góa, hoa tiên (tiêu), hóc, Huế”
- ✓ Chữ “**năm**” (year) => **niên**年(娘 浪 千 載 一 時
(Thương), hay là **tải** 載 Nàng rằng: “Thiên tải nhất thì)
- ❖ Mô hình Dịch máy: SMT, NMT, Transformer, T5, ... dựa trên
- ❑ tri thức ngôn ngữ và ngoài ngôn ngữ (extra-linguistic): rút ra từ
- ❑ **LLM** được huấn luyện trên **kho ngữ liệu** (corpus)
đúng: **lĩnh vực, thể loại, niên đại, vùng miền**
- ✓ có tính đến yếu tố **thời gian** (lịch sử) và **không gian** (địa lý)
- ✓ từ đồ thị tri thức (KG): **gia phả, ontology cây thuốc, địa danh, ...**

Polysemy of Nôm scripts

Tale of Kieu

冪 辭 沖 埃 得 些

Trăm năm trong cõi người ta

字 才 字 命 窖 羅 怙 饒

Chữ tài chữ mệnh khéo là ghét nhau

.....

浪 辭 嘉 靖 朝 明

Rằng: Năm Gia Tĩnh triều Minh

罊 方 滂 朗 叵 京 凭 傍

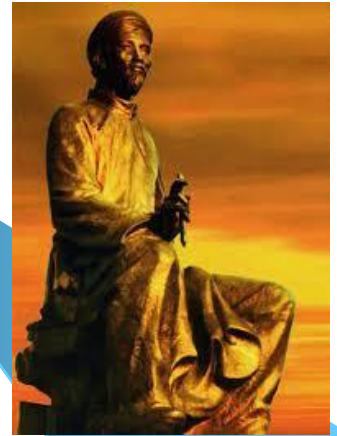
Bốn phương phẳng lặng hai kinh vững vàng

固 茹 員 外 戶 王

Có nhà viên ngoại họ Vương

家 資 擬 拱 常 常 壩 中

Gia tư nghị cũng thường thường bậc trung



Nguyễn Du
(1766-1820)

- *nghĩ (think)*
- *nghe (he)*

字 VĂN BẢN MỚI

TÀI LIỆU

CHỌN HÌNH ẢNH TẢI LÊN

Tự động

HÁN - NÔM



QUỐC NGỮ

嘯台閑事巖空
歪箕抱扒夕馱固身
扒風塵沛風塵
朱清高買特份清高
固兜偏為馱節
符才符命灘灘奇仁
固才麻泥之才
符才連貝符灾汶韻
包拏襪業匍身
拱脊責吝歪斯歪賒
善根於在惹些
符心箕買朋匹符才



ngắm thay muôn sự bởi trời
trời kia đã bắt làm người có thân
bất phong trần phải phong trần
cho thanh cao mới được phần thanh cao
có đâu thiên vị người nào
chữ tài chữ mệnh dồi dào cả hai
có tài mà cậy chi tài
chữ tài liền với chữ tai một vần
đã mang lấy nghiệp vào thân
cũng đừng trách lẫn trời gần trời xa
thiện căn ở tại lòng ta
chữ tâm kia mới bằng ba chữ tài

☐ Gỡ Hán-Nôm

122/500



☐ Hiện kết quả hình ảnh

Meaning Translation

汝等行看取敗虛
如何逆虜來侵犯
截然定分在天書
南國山河南帝居
南國山河

Nam quốc sơn hà

Nam quốc sơn hà Nam đế cư,
Tiệt nhiên định phận tại thiên thưⁱ.
Như hà nghịch lỗ lai xâm phạm,
Nhữ đẳng hành khan thủ bại hư.

Dịch nghĩa

Núi sông nước Nam thì vua Nam ở,
Cương giới đã ghi rành rành ở trong sách trời.
Cớ sao lũ giặc bạo ngược kia dám tới xâm phạm?
Chúng bay hãy chờ xem, thế nào cũng chuốc lấy bại vong.

Sự cần thiết của Dịch nghĩa

- **Dịch âm** (chuyên tự): khá hiệu quả với văn bản thuần Nôm, với các từ ngữ thuần Việt, dễ hiểu (vd: *trăm năm trong cõi người ta*)
- Tuy nhiên, với **văn bản Hán Nôm có chứa từ ngữ Hán Việt**
- => Cần phải **giải thích** (interpretation) mới hiểu đúng được.
- Vì **mức độ đồng âm** rất cao: do chữ Quốc ngữ là **chữ ghi âm**
- ✓ Một âm có nhiều nghĩa khác nhau: **nhập nhằng**
- ✓ Đồng âm giữa các từ thuần Việt và Hán Việt. Vd: *ba, cổ, trần,...*
- ✓ Di sản Hán Nôm là các văn bản cổ: **từ ngữ cổ, phong cách cổ.**
- ✓ Ngôn ngữ thay đổi qua các thời kỳ: **ngôn ngữ học lịch sử.**
- Nếu không giải thích (dịch nghĩa) => không hiểu được/hiểu sai!

Tri thức cho bài toán Dịch nghĩa

❖ Bài toán Dịch nghĩa đòi hỏi nhiều tri thức về:

▪ **Ngôn ngữ:**

- ✓ Vốn từ, thành ngữ, tục ngữ, quán ngữ, châm ngôn,...;
- ✓ Thể loại/lĩnh vực văn bản, khổ thơ, nhịp thơ;
- ✓ NNH lịch sử: từ nguyên, chiết tự, Hán/Việt cổ, từ địa phương;
- ✓ Sự khác biệt về từ ngữ tiếng Hán dùng ở VN và tiếng Hán ở TQ; tiếng Hán cổ và tiếng Hán hiện đại

▪ **Và ngoài ngôn ngữ:**

Nguyễn Phúc **Chu** => Phan **Châu** Trinh

Nguyễn Phúc **Cảnh** => cây **kiểng**

- ✓ **Lịch sử** (niên đại, triều đại, niên hiệu,...)
- ✓ **Văn hóa** (kỵ húy, điển tích, điển cố, dân gian, tôn giáo,...)
- ✓ **Địa lý** (vùng miền, nhân danh, địa danh thay đổi theo các thời kỳ)

Hà Nội: Thăng Long(1010), Đông Đô(1397), Đông Quan(1408), Đông Kinh(1427)

Tri thức cho bài toán Dịch nghĩa: **Ngữ âm lịch sử**

- “Nền xã tắc là nơi **báo bổ**, can chi mi đào lỗ đào hang”

[Nguyễn Đình Chiểu - thảo thử]

Từ cổ	Nghĩa
chác	<i>mua</i>
chí	<i>con tin</i>
tĩnh	<i>bàn thờ</i>
bao nả	<i>cao nhiều</i>
báo bổ	<i>báo đáp</i>

Ngữ âm học lịch sử:

Chữ Nôm cổ	Âm Việt cổ	Nghĩa
雷	<i>blời</i>	<i>trời</i>
𩇛	<i>tlâm</i>	<i>thâm</i>
𩇛	<i>kjàu</i>	<i>giàu</i>

STT	Âm	Chữ cũ	Tần số	Chữ mới	Tần số
1.	Một	蔑	4	沒	2
2.	Buồn	盆	1	愔	0
3.	Đêm	店	16	𡇗	0
4.	Vẻ	尾	2	𡇗	0
5.	Xưa	初	9	𡇗	0
6.	Đến	旦	5	𡇗	0
7.	Mặt	𡇗	11	𡇗	0
8.	Chân	眞	6	𡇗	0
9.	Con	昆	40	𡇗	0
10.	Mũi	每	2	𡇗	0
11.	Thấy	体	31	𡇗	3
12.	Ra	𡇗 / 囉	33	𡇗	36

[Trần Trọng Dương – Nghiên cứu chữ Nôm lịch đại]

Tri thức cho Dịch nghĩa: Khác biệt từ Hán VN & TQ

- ❑ Một số từ Hán Việt của VN khác nghĩa với của TQ
- ❑ Một số từ Hán Việt do VN tự sáng tạo

Từ Hán (VN)	Từ Hán TQ	
Tiến sĩ	Bác sĩ	
Kỹ sư	Công trình sư	
Sinh học	Sinh vật	
	Khẩu thủy	Nước miếng

Từ Hán (Việt)	Nghĩa VN	Nghĩa TQ
Hạ thủ	Xuống tay, Sát hại	Bắt tay làm
Đình ninh	Yên trí	Dặn dò
Mê ly	Rất hay	Mơ hồ
Dã tâm	Lòng dạ hiểm ác	Tham vọng
Thủ đoạn	Mưu mẹo	Phương pháp
Bồi hồi	lo lắng, bồn chồn	Đi đi lại lại

Tri thức cho Dịch nghĩa: Từ/ngữ Hán Việt cổ định

- ❑ Nhận diện một số từ Hán Việt mà không cần diễn Nôm, vd:

梅 骨 格 雪 精 神

熱血: nhiệt huyết !=> *máu nóng*

Mai **cốt cách** tuyết tinh thần

- Nhận diện ngữ cổ định: thành ngữ, quán ngữ, điển tích,
an cư lạc nghiệp, án binh bất động, hào hoa phong nhã
- Tri âm (Chung Tử Kỳ, Du Bá Nha), hàn thực (Giới Tử Thôi),...
- Nhận diện thực thể có tên (NER): *Per-Loc-Org-Time-Title*,...
- 雄王: Hùng Vương !=> *vua khỏe*
- 昇龍: Thăng Long !=> *rồng bay*

Xây dựng ngữ liệu huấn luyện cho Dịch nghĩa

除夕其四 Trừ tịch kỳ 4 Đêm trừ tịch kỳ 4	Nguyễn Khuyến	書齊不寐獨沉吟 滴滴銅龍夜轉深 遠近競推除夕鼓 寒喧誰記昔年心 盧陵悶甚憐蟲響 林仲間多學鷓鴣 坐待明朝是元旦 改觀萬像一橫襟	thư trai bất寐 độc trầm ngâm trích trích đồng long dạ chuyển thâm viễn cận cạnh thôi trừ tịch cổ hàn huyền thủy ký tích niên tâm lư lăng muộn thậm liên trùng hưởng lâm trọng nhàn đa học khẩu âm toạ đãi minh triều thị nguyên đán cải quan vạn tượng nhất hoành khâm	trong phòng sách một mình không ngủ ngồi trầm ngâm đồng hồ nước điểm từng giọt đêm đã khuya xa gần rộn lên tiếng trống thúc giao thừa lúc hàn huyền ai nhớ được nỗi lòng những năm xưa lư lăng buồn quá thương giun dế kêu lâm trọng nhàn hạ học tiếng chim non hót ngồi đợi xem sáng mai là tết nguyên đán chỉ mới xoay ngang vạt áo một cái mà mọi cảnh vật đổi thay
白藤江 Bạch Đằng giang Sông Bạch Đằng	Nguyễn Sưởng	鯨觀如山草木春 海潮洶洶石嶙峋 誰知萬古重興業 半在關河半在人	kinh quán sơn hà thảo mộc xuân hải triều húng húng thạch lân tân thủy tri vạn cổ trùng hưng nghiệp bán tại quan hà bán tại nhân	mồ chôn quân thù cao như núi cây cỏ xanh tươi nước thủy triều ngoài biển ầm ầm đá núi lởm chởm mấy ai biết sự nghiệp muôn thuở đời trùng hưng một nửa nhờ địa thế núi sông một nửa do con người

Dịch âm và Dịch nghĩa văn bản Hán cổ



Sản phẩm này đã có mặt trên website:
tools.clc.hcmus.edu.vn hay kimtudien.com.vn
và trên các cửa hàng ứng dụng App Store và
Google Play với tên gọi là "Kim Hán Nôm".



HÁN - NÔM



QUỐC NGỮ

獄中無酒亦無花
對此良宵奈若何
人向窗前看明月
月從窗隙看詩家



ngục trung vô tửu diệc vô hoa
đối thử lương tiêu nại nhược hà
nhân hướng song tiền khan minh nguyệt
nguyệt tòng song khấp khán thi gia



Dịch nghĩa (thử nghiệm):

trong tù không rượu cũng không hoa
trước cảnh đẹp đêm nay biết làm thế nào
người hướng ra trước song ngắm trăng sáng
từ ngoài khe cửa trăng ngắm nhà thơ

31 / 500



ALIGNMENT (dóng hàng):

- MED (Minimum Edit Distance): Levenshtein

1. Character alignment:

Two strings and their **alignment**:

I	N	T	E	*	N	T	I	O	N
*	E	X	E	C	U	T	I	O	N

2. Word alignment:
3. Inflected word alignment
4. Transliteration word alignment
5. Synonym word alignment
6. Translation word alignment
7. Translation sentence alignment
8. Comparable meaning sentence alignment
9. Related discourse alignment (Q&A)

ALIGNMENT EXTENSION

1. “intention” vs. “execution”
 2. “I am a teacher” vs. “I am not a student”
 3. “I buy one box” vs. “He bought two boxes”
 4. “trăm năm trong cõi người ta” vs “稊𦵏𦵏𦵏𦵏𦵏些”
 5. “You are a well-known teacher” vs. “He is also a famous professor”
 6. “I am a new teacher” vs “Tôi là một giáo_viên mới”
 7. Paragraph/text alignment
 8. “We are teachers” vs “Mình cũng đều là nhà giáo cả”
 9. “Input:Question/Request” vs. “Output: Answer/Respond”
- E.g: OCR: img => char; VR: voice => txt; etc.

遷都詔

昔商家至盤庚五遷，周室逮成王三徙。豈三代之數君俱徇己私，妄自遷徙，以其宅中圖大，為億萬世子孫之計。上謹天命，下因民願，苟有便輒改，故國祚延長，風俗富阜。而丁黎二氏，乃徇己私，忽天命，罔蹈商周之迹，常安厥邑于茲，致使世代弗長，算數短促，百姓耗損，萬物失宜。朕甚痛之，不得不徙。

羣臣勸朕，朕亦
 思之，彼嗇斯豐
 浪蕩，嘉靖朝明
 固茹員外戶王
 沒跡，猥次卒，悉
 頭悉，仝，素娥
 枚骨格，雪精神
 雲祐莊重，恪為
 花琪王說端莊
 翻強色稍漫，亦

符才，符命，害，哭，怙，饒
 仍調，龔，覽，也，勃，疸，悉
 丞，撐，涓，貝，鴈，紅，打，煙
 風情，固，錄，群，傳，史，撐
 眾方，榜，潮，仝，京，凭，鑽
 家資，擬，拱，常，又，堀，中
 王兒，罷，符，綏，泐，儒，家
 翠，翹，罷，姊，掩，罷，翠，雲
 沒，馱，沒，跡，進，分，院，進
 困，腹，苔，憚，惶，蟬，安，囊
 遷，輸，諾，還，雪，讓，牟，腹
 搗，皮，才，色，吏，罷，分，欣



積

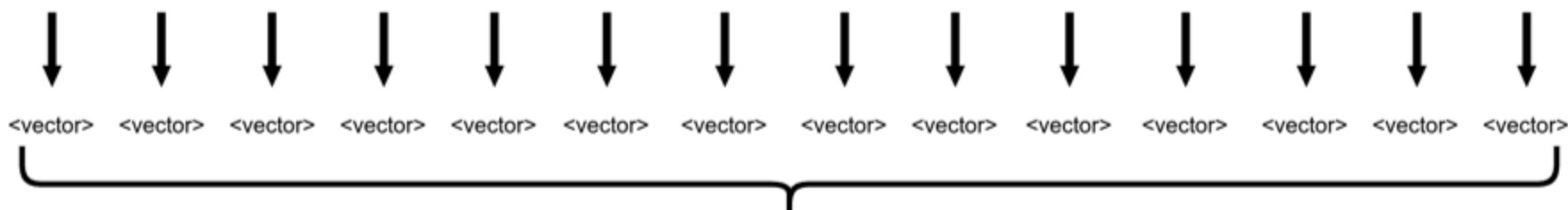
積

1 2 3 4 5
𩇛 禾 喜 𩇛 貝

6 7 8

9 10 11
忍 罪 苦

12 13 14
ト ㄥ マ



Transformation Based Learning (TBL)

